

// MASTER INVESTOR BRIEFING (2026)

The Operating System for Military-Grade Robotics in GPS-Denied Theatres

Hardware-agnostic, zero-external-dependency 3D SLAM, kinematic trajectory planning, and AI threat detection software for Unmanned Ground Vehicles (UGVs).

1,276 / 1,276

TESTS PASSED (100%)

<15ms

HYBRID A* PLANNING

STANAG 4586

NATO ARCHITECTURE

0%

FOREIGN BLACK-BOX IP

// CRITICAL DEFENSE DEFICIENCIES

Electronic Warfare Has Broken Legacy Autonomous Robotics

Modern battlefield environments expose three fundamental architectural vulnerabilities in current UGV fleets:

1. GPS-Denied Deadlocks

Modern EW and anti-satellite jamming render satellite navigation completely useless within seconds, stranding non-autonomous robotic platforms.

2. Lethal Asymmetric Traps

Anti-Tank/Personnel mines, tripwires, and IEDs cause over 60% of ground vehicle casualties before reaching tactical mission objectives.

3. Software Bottlenecks

Over 90% of vehicle chassis manufacturers build exceptional mechanical platforms, but lack proprietary autonomy, SLAM, and AI detection software stacks.

// THE TRUSTIA AUTONOMY STACK

Full-Stack Sovereign Autonomy for Unmanned Fleets

A modular software engine that turns any mechanical chassis into an intelligent tactical combat platform in under 10 minutes.

GPS-Denied 3D SLAM

Centimeter-accurate 3D LiDAR, Visual Odometry, and IMU pose-graph optimization under active jamming.

Hybrid A* Kinematic Planner

Sub-15ms real-time pathfinding respecting turning radii, slip angles, and severe terrain gradients.

AI Threat Fusion

Ground-penetrating radar and thermal anomaly fusion creating automated 30m hazard quarantine rings.

Plug & Play HAL

Native CAN FD, ROS 2, and SAE AS6091 (JAUS) interfaces compatible with wheeled and tracked platforms.

// EMBARGO-PROOF INDEPENDENCE

100% Sovereign Code Verified by 1,276 Automated Tests

Zero foreign library dependencies, eliminating backdoor risks and international export licensing vulnerabilities.

1,276 Automated Tests

100% test pass rate across unit algorithms, mathematical convergence, and extreme climate stress matrices.

Zero-Failure Simulation

50/50 consecutive autonomous missions completed with zero collisions across 4 distinct harsh operating environments.

Cryptographic Security

Hardware-level HMAC-SHA256 telemetry authentication and automatic LinkLoss Return-to-Home (RTH) failsafes.

// END-TO-END DEFENSE ARCHITECTURE

From Low-Level Actuation to Multi-Vehicle Swarm C2

1. Trustia Core Middleware

Modular perception, 3D Pose Graph SLAM, kinematic trajectory planner, and Stanley/Pure Pursuit vehicle controllers.

2. AI Threat Fusion Suite

GPR radar fusion, thermal anomaly analysis, and 30m automatic hazard buffer generation for IEDs and CBRN plumes.

3. Air-Ground Swarm Engine

Multi-agent consensus in Wedge, Line, Column, and Diamond tactical formations with autonomous UAV-to-UGV target handoff.

4. Tactical C2 Command Console

MIL-STD-2525 military map GUI for live fleet telemetry, mission dispatch, and instant hardware E-Stop.

// HIGH-MARGIN SOFTWARE RECURRING REVENUE

Scalable SaaS & Defense Integration Model

Tier 1: OEM Vehicle License**\$15,000 – \$35,000**

Per vehicle / year. Sold directly to UGV manufacturers and defense contractors per active platform.

Tier 2: Fleet & Swarm C2**\$50,000 / Center / Yr**

Enterprise command license for multi-robot base stations, tactical mission dispatch, and C2 servers.

Tier 3: Custom NRE Contracts**\$250K – \$1.0M**

Bespoke military vehicle integration, custom sensor payloads, and government field qualifications.

// \$34.8B DEFENSE ROBOTICS BOOM

Massive Secular Tailwinds in Unmanned Warfare

\$34.8 Billion

TAM: GLOBAL DEFENSE ROBOTICS (2030)

\$8.2 Billion

SAM: UGV AUTONOMY & GPS-DENIED NAV

\$450 Million

SOM: INITIAL NATO & ALLIED TARGET

// MARKET COMPARISON MATRIX

Why Trustia Dominates Legacy & Open-Source Solutions

CAPABILITY MATRIX	LEGACY DEFENSE (LOCKHEED, GD)	OPEN SOURCE (ROS NAV2, AUTOWARE)	TRUSTIA AI
GPS-Denied Precision	Costly (\$500K+ INS Hardware)	Severe Drift / Unstable	Centimeter 3D SLAM (\$0 GPS)
Mine / IED AI Quarantine	Separate Manual Sensor Kits	None Available	Native 30m Auto Quarantine Ring
Integration Agility	Proprietary Lock-in (2-3 Years)	Fragile Dependency Hell	Plug & Play (CAN FD / ROS 2 / JAUS)
Business Model	Multi-Million Cost-Plus	No SLAs / Uncertified	Predictable High-Margin OEM SaaS
Verification Rigor	Black-Box Proprietary	Untested in Battlefield	1,276 Automated Verified Tests (100%)

// LEADERSHIP & TECHNICAL EXCELLENCE

Executive & Autonomous Systems Leadership

Murat Furkan Bayram**Founder & Lead Autonomous Systems Engineer**

Certified Advanced Entrepreneur (KOSGEB) and Defense Industry Academy autonomy engineer. Architected Trustia's sovereign core from scratch, mastering GPS-denied 3D Pose Graph SLAM, kinematic Hybrid A* planning, and multi-robot swarm consensus.

Institutional Credentials

- ✓ KOSGEB Advanced Entrepreneurship Certified #2026
- ✓ BTK & SSB Defense Industry Academy Certified
- ✓ Delaware C-Corporation Legal Shield (USA)
- ✓ Istanbul Defense R&D & Field Testing Facility

// STRATEGIC SEED ROUND

Building the Future of Autonomous Defense

Accelerating military field trials, defense OEM integrations, and global expansion across NATO and allied defense contractors.

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